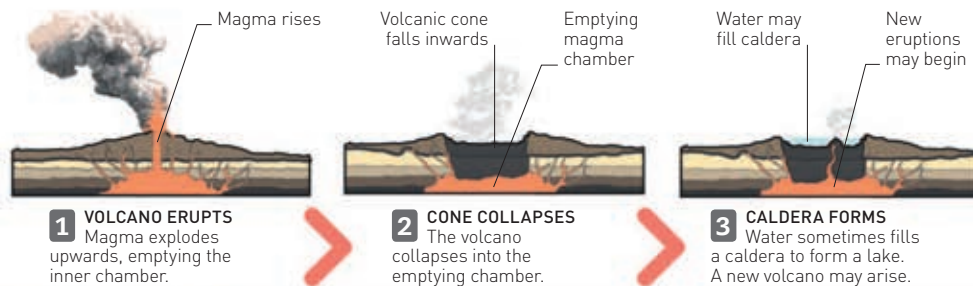


CALDERAS

A caldera is a vast bowl in the ground. Calderas form when a volcano comes apart during an eruption, and the surface collapses into the emptying magma chamber.



A THICK LAVA FLOW CAN TAKE YEARS TO COOL DOWN COMPLETELY

Magma erupts from surface crack

1,200°C (2,190°F)
The temperature at Earth's surface of the hottest lavas.

600°C (1,112°F)
The flame temperature of a lighted match.

100°C (212°F)
The boiling point of water at sea level.

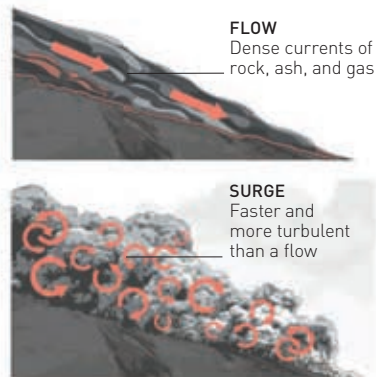
RED-HOT

Scientists are exploring ways of tapping into the huge energy produced by hot magma. One day, this could provide the world with a big new source of power.

Chamber full of molten rock, or magma

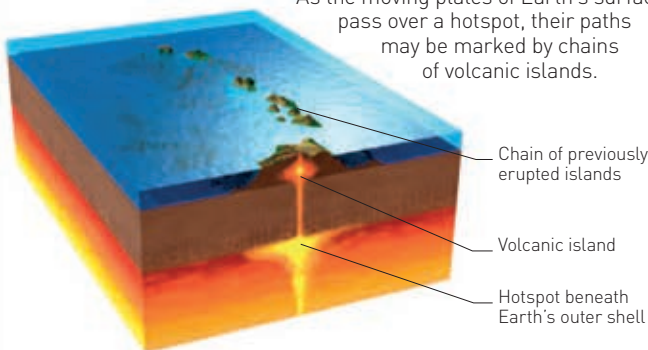
FLOWS AND SURGES

Pyroclastic flows are lethal currents of hot gas, ash, and rocks. These currents race down a volcano, destroying everything in their path. Just as deadly are billowing, choking clouds called pyroclastic surges. They contain more gas than pyroclastic flows and can move faster.



VOLCANIC ISLAND CHAINS

Beneath some ocean floors are volcanic areas, or "hotspots". As the moving plates of Earth's surface pass over a hotspot, their paths may be marked by chains of volcanic islands.



LIVING NEAR A VOLCANO

Many people spend their lives next door to a volcano. They are prepared to put up with the risk of danger because there are a few advantages, too.

POSITIVES



TOURISM
Sightseers bring in money.



AGRICULTURE
Volcanic soil is good for growing crops.



ENERGY
Hot underground water is used by industry.

NEGATIVES



DEATH
Eruptions can kill people and ruin land.



MUDFLOWS
Violent floods wash down volcanic debris.



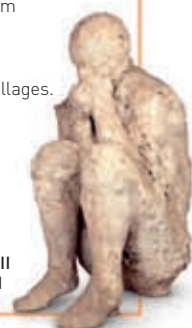
REFUGEES
People lose their homes and livelihoods.



MOST DEADLY

These are 10 of the biggest volcanic disasters. The worst, Tambora, killed more than 70,000 people. The most famous is Vesuvius's eruption in 79 CE.

- 1 MOUNT TAMBORA, 1815**
Sumbawa island, Indonesia. Ejected ash blocked the sun and lowered global temperatures.
- 2 KRAKATOA, 1883**
Krakatoa island, Indonesia. Made the loudest bang ever recorded and blew up most of the island.
- 3 MT PELEE, 1902**
Martinique, Caribbean Islands. Ash and gas flowed at speeds of more than 600 kph (370 mph).
- 4 NEVADO DEL RUIZ, 1985**
Colombia. Gigantic mudflows overwhelmed an entire town.
- 5 MOUNT UNZEN, 1792**
Japan. Created a landslide and a tsunami.
- 6 LAKI, 1783**
Iceland. Poisonous gas killed half of Iceland's farm livestock.
- 7 KELUT, 1919**
Java, Indonesia. Mudslides destroyed more than 100 villages.
- 8 SANTA MARIA, 1902**
Guatemala. Ash detected 4,000 km (2,500 miles) away.
- 9 GALUNGGUNG, 1882**
Java, Indonesia. Destroyed 114 villages.
- 10 VESUVIUS, 79 CE**
Italy. The cities of Herculaneum and Pompeii were wiped out. An eruption in 1631 caused further deaths.



CAST OF A POMPEII DISASTER VICTIM

SUPERVOLCANOES

These are the monsters, capable of eruptions thousands of times larger than those of any other kind of volcano. Luckily, there aren't many of them. Here are some of the most important.

YELLOWSTONE
This map shows the vast area of North America affected in one of Yellowstone's ancient eruptions.



Yellowstone caldera

- YELLOWSTONE CALDERA**
Wyoming, USA. Makes up much of Yellowstone Park.
- LONG VALLEY CALDERA**
California, USA. Recent uplifting of ground observed.
- VALLES CALDERA**
New Mexico. Hot springs are a sign of volcanic activity.
- LAKE TOBA**
Sumatra, Indonesia. World's largest volcanic lake.
- LAKE TAUPU**
New Zealand. Has erupted 28 times.
- AIRA CALDERA**
Japan. Contains a currently active volcanic cone.

WHERE IN THE WORLD?

Volcanoes emerge in clusters in just a few places around the world. There are large numbers in the area called the "Ring of Fire" that circles the Pacific Ocean. Iceland, East Africa, and the Caribbean are big volcanic regions, too.



IN SPACE

Earth is not the only body in the Solar System to have volcanoes. Some of our neighbours in space have many volcanic regions.



TRITON
Volcanic eruptions on Neptune's biggest moon can last a whole year.



IO
This moon of Jupiter has many active volcanoes.

OLYMPUS MONS

MT EVEREST



SLEEPING GIANT
Olympus Mons, a vast volcano on Mars, is three times as high as Mount Everest. It is not active at the moment.

VENUS
The planet's surface is made almost entirely of volcanoes.

